

Q1. What is a Webpage?

Ans: A webpage is a document written in HTML and can be viewed on any web browser. It is contained within the web server, which can be accessed by entering the URL for that web page, and once it is loaded, it appears on the user's web browser. Each webpage is linked with a unique URL; hence two pages cannot have the same URL.

Basic terms that are used with Webpage:

- **WebSite:** A website is a collection of several web pages. These pages are linked together with hyperlinks. A website has a unique domain name, and we can access it by entering that domain name in the URL.
- **Search Engine:** A search engine is an internet service that helps users find any information available on the internet. Some examples of search engines are **Google, Yahoo, Bing**, It is usually accessed with the help of Web browser.
- **Web Browser:** A web browser or simply browser is application software used to access the internet. Some examples of Web browsers are **Google Chrome, Microsoft Internet Explorer, Safari, etc.** It does two things:
 - It connects to a web server on the internet and requests a page that the user wants to view; once it finds that page, it displays it on its device.
 - It can interpret the set of HTML tags within a page to display the page in the correct format.
- **Webserver:** A web server can be understood as a computer that hosts or provide a website on the internet. It contains webserver software and component files of a website such as **HTML document, images, CSS stylesheet, and JS files.**

Q2. Explain Characteristics of a Web Page

Ans:

Following are some characteristics of a Web page:

- A simple webpage can be created very quickly.
- It takes very little time to create a webpage compared to a Website.
- A web page and a website should be compatible with any device, such as Mobile, Desktop, Laptop, etc.
- The search engine provides a web page through a link, and when a user clicks on that link, it is redirected to the webpage of a website.
- A webpage can have any type of information including videos, and audios.
- It can be made up of only HTML (Hypertext Markup Language), or CSS, or JavaScript for dynamic and attractive behavior.

Website	Webpage
A website is a collection of different web pages that are linked together with hyperlinks.	A webpage is a single hypertext document.
It consists of more than one webpage.	It is a single document that is displayed on the user's browser.
To develop a website, developers need more skills and more time compared to a webpage.	To develop a webpage, developers need basic HTML knowledge and less time.
A website is accessed through its domain name, and it does not include any extension in the URL.	A webpage is accessed through a unique URL with some extension.
It can contain information for different entities, such as Javatpoint.com, which contains information about different technologies.	It can contain information for a single entity, such as currently viewing a web page containing information about this page only.
It is a little challenging to create a perfect website and requires lots of programming.	It is very simple to create a webpage.
Some examples of the website are Javatpoint.com, Amazon.com, etc.	Some examples of Webpages are the currently viewing page, contact page, registration page, the home page, etc.

Elements of a Webpage

The main element of the webpage is a text file composed of HTML. Apart from this, a webpage can also have the following elements:

- **CSS:** The CSS code is used to make the page more interactive and control its look and feel.
- **Scripts:** The JavaScript code is included in a webpage to add interactivity to the page and add more functionalities to it.
- **Media:** It is used to include media components such as audio, video, and images

Q: Explain Web Browser

Ans: A software application used to access information on the World Wide Web is called a Web Browser. When a user requests some information, the web browser fetches the data from a web server and then displays the webpage on the user's screen.

Q: Brief: History of Web Browser

Ans:

Today web browsers are easily accessible and can be used on devices like computer, laptops, mobile phones, etc. but this evolution of making browsers available for easy use took many years.

Given below are some salient points which one must know with regard to the history of web browsers:

- **“WorldWideWeb”** was the first web browser created by Tim Berners Lee in 1990. This is completely different from the World Wide Web we use today

- In 1993, the “**Mosaic**” web browser was released. It had the feature of adding images and an innovative graphical interface. It was the “the world’s first popular browser”
- After this, in 1994, Marc Andreessen (leader of Mosaic Team) started working on a new web browser, which was released and was named “**Netscape Navigator**”
- In 1995, “**Internet Explorer**” was launched by Microsoft. It soon overtook as the most popular web browser
- In 2002, “**Mozilla Firefox**” was introduced which was equally as competent as Internet Explorer
- Apple too launched a web browser in the year 2003 and named it “**Safari**”. This browser is commonly used in Apple devices only and not popular with other devices
- Finally, in the year 2008, Google released “Chrome” and within a time span of 3 years it took over all the other existing browsers and is one of the most commonly used web browsers across the world

Q: Give of Web Browser

Ans: The functions of all web browsers are the same. Thus, more than the different types there are different web browsers which have been used over the years.

Discussed below are different web browser examples and their specific features:

1. WorldWideWeb

- The first web browser ever
- Launched in 1990
- It was later named “Nexus” to avoid any confusion with the World Wide Web
- Had the very basic features and less interactive in terms of graphical interface
- Did not have the feature of bookmark

2. Mosaic

- It was launched in 1993
- The second web browser which was launched
- Had a better graphical interface. Images, text and graphics could all be integrated
- It was developed at the National Center for Supercomputing Applications
- The team which was responsible for creating Mosaic was lead by Marc Andreessen
- It was named “the world’s first popular browser”

3. Netscape Navigator

- It was released in 1994
- In the 1990s, it was the dominant browser in terms of usage share
- More versions of this browser were launched by Netscape

- It had an advanced licensing scheme and allowed free usage for non-commercial purposes

4. Internet Explorer

- It was launched in 1995 by Microsoft
- By 2003, it has attained almost 95% of usage share and had become the most popular browsers of all
- Close to 10 versions of Internet Explorer were released by Microsoft and were updated gradually
- It was included in the Microsoft Windows operating system
- In 2015, it was replaced with “Microsoft Edge”, as it became the default browser on Windows 10

5. Firefox

- It was introduced in 2002 and was developed by Mozilla Foundation
- Firefox overtook the usage share from Internet Explorer and became the dominant browser during 2003-04
- Location-aware browsing was made available with Firefox
- This browser was also made available for mobile phones, tablets, etc.

6. Google Chrome

- It was launched in 2008 by Google
- It is a cross-platform web browser
- Multiple features from old browsers were amalgamated to form better and newer features
- To save computers from malware, Google developed the ad-blocking feature to keep the user data safe and secure
- Incognito mode is provided where private searching is available where no cookies or history is saved
- Till date, it has the best user interface

Types of a Web page

There are mainly two types of a Web page based on functionality:

- Static Webpage
- Dynamic Webpage

Static Webpage

Static webpages are those webpages that cannot be modified or altered by the client. These are also known as stationary or flat web pages. They are displayed on the client's browser in the same format and manner as they are saved in the webserver. Users can only load the page and read the information but cannot perform any change on the page. A static webpage is generally made up of HTML and CSS only.

Dynamic Webpage

As the name suggests, Dynamic webpages are dynamic, which means it shows different information at different point of time. The dynamic webpage shows different content each time it is viewed. There are two types of Dynamic web pages, which are:

- **Server-Side Dynamic Webpage:** These web pages are created using Server-side scripting. These pages are changed when they are visited or viewed. Some examples of server-side pages are **login pages, submission forms, shopping carts, etc.** Various scripting languages such as **PHP, ASP.Net, JSP, etc.**, can be used for server-side scripting.
- **Client-side Dynamic Webpage:** These web pages are created using Client-side scripting. These pages get changed in response to actions that occur within that page, such as mouse or keyboard action. Scripting languages such as **JavaScript, Dart, etc.**, can be used for client-Side scripting

Q2. Why HTML is important for Web Development? Which other languages are support for web site development.

Ans: HTML is primarily used for creating web pages. HTML is free to use and ensures your text, images, and other elements are displayed as intended. With HTML, not only can you add headings, paragraphs, lists, and other elements to your page — you can also embed images, videos, audio files, and other multimedia. Web Development is working with a combination of both frontend and backend languages that are used by developers and programmers to build a well-functioning website or web application.

- Top 4 Backend Web Development Languages
 1. Python
 2. PHP
 3. Java
 4. C#
- Top 4 Frontend Web Development Languages
 1. JavaScript
 2. React
 3. Angular
 4. vueJS

Q3. In case a programmer writes comments in the code, what will be impact on the output? Explain comments types in HTML with example.

Ans: The comments are not compiled or considered an integral part of the program by the compiler, and hence they do not affect any behavior or functionality of the program. Comments are solely intended for human users and developers for their understanding and maintenance of the code. Comments can be used to hide content Single Line.

```
<!-- This is a comment -->
<p>This is a paragraph.</p>
<!-- Remember to add more information here -->
```

You can also hide more than one line. Everything between the `<!--` and the `-->` will be

hidden from the display.

```
<p>This is a paragraph.</p>  
<!--  
<p>Look at this cool image:</p>  
-->  
<p>This is a paragraph too.</p>
```

Hide Inline Content

Comments can be used to hide parts in the middle of the HTML code.

```
<p>This <!-- great text --> is a paragraph.</p>
```